

My Presentation Script

Loan Approval and Risk Assessment System — What I'll Say to My Teacher

This is written exactly the way I would explain my project out loud, in my own words, start to finish. The green boxes are things to actually say. The small italic notes underneath are just tips for me, not things to say out loud. I can read this almost word-for-word, or use it as a guide and say it more naturally in my own voice.

1. How I'll Open

"Good morning/afternoon. My project is called the **Loan Approval and Risk Assessment System**. The idea is simple — banks get thousands of loan applications, and they need a fast, consistent way to predict whether an applicant is likely to repay a loan or not. So I built a machine learning model that looks at an applicant's details — like their income, credit history, and loan amount — and predicts whether their loan should be approved or rejected."

■ *Say this part slowly and clearly — it's the one sentence that tells the teacher exactly what my project does, so I want it to land.*

2. Explaining What Machine Learning Is (in case they ask)

"Normally, when we code something, we write the exact rules ourselves — like 'if income is above this number, approve the loan.' But with machine learning, I don't write the rules. Instead, I give the computer hundreds of real past examples of loan applications, along with whether they were actually approved or rejected, and the computer **learns the rules by itself** by finding patterns in that data."

3. "Did You Build This Model, or Is It Already Made?"

This is a question I should be ready for, and I want to be precise about it so I sound confident, not confused.

"Random Forest itself is a well-known machine learning algorithm — it's not something I invented, it already exists and is built into a Python library called scikit-learn. But the **model** — meaning the actual trained result — is something I built myself, from scratch, using my own dataset. Nobody had trained this exact model before. So I used an existing algorithm, but I did the training myself."

■ *If they push further and ask 'so what did you actually do then?' — say I collected the dataset, cleaned it, chose the settings, trained it, and tested it. That's the real work, even though the algorithm existed before.*

4. Explaining Random Forest

"A Random Forest is made up of many decision trees. A decision tree is basically a flowchart of yes/no questions — like 'Is the applicant's credit history good? If yes, is their income above a certain amount?' and so on, until it reaches a final decision. One tree on its own can make mistakes or be biased toward the exact data it was trained on. So instead of trusting one tree, I trained **200 different trees**, each looking at a random slice of the data. When a new applicant comes in, all 200 trees vote, and whichever answer gets the most votes — approve or reject — becomes the final prediction. It's basically the 'wisdom of the crowd' applied to decision-making."

5. Explaining My Dataset

"I used a dataset of **491 real loan applications**. Each one has details like the applicant's income, whether they're married, their education level, their credit history, the loan amount they requested, and the property area — and each one also has the actual outcome, approved or rejected. This is a well-known dataset that's commonly used in loan-prediction projects, so it's a credible choice for learning and demonstrating this kind of system."

"Some of the data was messy — a few applicants had missing information, like a missing credit history value. I handled that by filling in missing numbers with the median value, and missing categories with the most common value, which are standard, safe techniques for cleaning data."

6. Explaining How I Trained and Tested It

"I split my data into two parts — 80% for training, which is 392 applications, and 20% for testing, which is 99 applications. The model only learns from the training data. Then I test it on the remaining 99 applications that it has **never seen before**, to check whether it actually learned useful patterns, or just memorized the training data."

■ *If asked why this split matters: it's the only honest way to measure performance — testing on data the model already saw would make the results look artificially good.*

7. Presenting My Results

"On the unseen test data, my model achieved **81.8% accuracy** — meaning out of 100 applications, it correctly predicts about 82 of them. I also looked at other metrics: precision of about 85%, meaning when the model says 'approved,' it's right 85% of the time; and recall of about 90%, meaning it correctly identifies 90% of applicants who should truly be approved. I also calculated the ROC-AUC score, which came out to 83.2% — this measures how well the model separates good applicants from risky ones, where 100% would be perfect and 50% would be random guessing."

Metric	My Score
Accuracy	81.8%
Precision	84.9%
Recall	89.9%
F1 Score	87.3%
ROC-AUC	83.2%

■ *Have this table memorized or written on a small card — it's the numbers I'm most likely to get asked to repeat.*

8. Explaining the Feature Importance Chart

"I also analyzed which factors the model relies on most when making decisions. The most important factor by far was **credit history** — more than income, more than loan amount. This actually makes a lot of sense, because that's exactly what real banks care about most too. To me, this was a good sign that the model learned real, meaningful patterns instead of just noise in the data."

9. What I'll Say During the Live Demo

"Now I'll run the model live. [Run: `python loan_model.py`] You can see it going through each step — loading the data, cleaning it, training the 200 trees, and then evaluating on the test set. At the end, it takes one sample applicant from the test set and shows a real prediction, along with a confidence percentage — this is the model actually making a decision in real time."

■ *If the live run fails for any reason (no internet, wrong folder, etc.), fall back calmly: 'I also have the results and charts saved from a previous run,' and show the saved PNGs and results_summary.txt instead. No need to panic.*

10. How I'll Close

"This is where I am for this milestone — a working, trained Random Forest model with about 82% accuracy on real loan data. For the final version, I'm planning to tune the model's settings further to improve accuracy, compare it against other algorithms like Logistic Regression to justify why Random Forest is the best choice, and possibly build a simple web form so a user can type in an applicant's details and get a live prediction. That's my project so far — happy to answer any questions."

Practice this once out loud tonight, even just to myself. I've got this.